Materials Science & Engineering A 861 (2022) 144324

Contents lists available at ScienceDirect

Materials Science & Engineering A

journal homepage: www.elsevier.com/locate/msea

Effect of cold rolling and subsequent heat treatment on microstructural evolution and mechanical properties of Fe-Mn-Al-C-(Ni) based austenitic low-density steels

Bidyapati Mishra [a,b,**], Rajdeep Sarkar [a], Vajinder Singh [a], Deepak Kumar [a], A. Mukhopadhyay [a], V. Madhu [a], M.J.N.V. Prasad [b,*]

[a] Defence Metallurgical Research Laboratory, Hyderabad, 500058, India
[b] Microstructural Engineering and Mechanical Performance Lab, Department of Metallurgical Engineering and Materials Science, Indian Institute of Technology Bombay, Mumbai, 400076, India

ARTICLE INFO

Keywords:
Low-density steels
Austenite
Cold rolling
Microstructure
Tensile properties

ABSTRACT

Thermo-mechanical processing in a two-phase domain of low-density steel (LDS) has the potential of enhancing strength-ductility combination. In the present study, the effects of alloy composition, cold rolling and heat treatment temperature (973–1173 K) on concurrent precipitation of κ-carbide/B2 particles and recrystallization and subsequently, mechanical properties include hardness variation and tensile response are evaluated. In order to understand the synergistic effect of Mn/Ni ratio towards selecting the two-phase processing window, three LDS alloy compositions, namely Fe–28Mn–9Al-0.9C, Fe–28Mn–9Al-0.9C–5Ni and Fe–15Mn–9Al-0.9C–5Ni (in weight percent) are vacuum induction melted, hot rolled and then cold rolled to 50% reduction in thickness at room temperature. The LDS alloys containing Ni showed much delayed recrystallization as compared to Ni-free LDS. The Mn/Ni ratio played a vital role towards controlling precipitation and recrystallization and accordingly, the mechanical properties. All the three alloys exhibited similar hardness, strength and ductility levels after 50% cold reduction. Upon heat treatment, there was precipitation of intergranular and intragranular B2 phase particles along with κ-carbides, whose fraction decreased with increasing the temperature. The tensile response of LDS alloys was influenced primarily by the volume fraction of these precipitates as well as the recrystallized austenite grains. Planar slip activity, dynamic slip band refinement in conjunction with dislocation-twin interaction mechanisms were noticed in fully recrystallized austenite and thereby leading to enhanced strain hardening and ductility. Furthermore, lowering Mn in Ni containing LDS reduced the processing temperature window of the alloy in two-phase region.

1. Introduction

Austenite based low-density steel (LDS) is one of the advanced high strength steels and is considered as a potential candidate material for certain automotive and defense applications owing to its high specific strength and ductility combinations. These alloys typically have high Mn and C content to stabilize austenite phase at room temperature. Apart from inducing significant hardening by solid solution and facilitating intermetallic precipitation, aluminum (Al) addition is known to reduce physical density of steel significantly [1–4]. Further, nickel (Ni) addition to Fe–Mn–Al–C based alloy system leads to formation of hard intermetallic particles such as Fe(Ni)Al (B2) and $(Fe,Ni)_3Al$ ($D0_3$), and thereby attain superior strength in LDS alloys [5–10]. Based on alloying and processing conditions, Fe-(25–30)Mn-(8–11)Al-(0.8–1.2)C-(0–5)Ni (all elements in wt. %) based LDS alloys show triplex structure with austenite (γ), ferrite (disordered/ordered B2) and kappa (κ) carbides as main microstructural constituents [4–10]. Most of these alloys are reported to be processed at temperatures higher than 1173 K, having completely recrystallized austenite as predominant microstructure, resulting into lower yield strengths. Strength enhancement is achieved

* Corresponding author.
** Corresponding author. Defence Metallurgical Research Laboratory, Hyderabad, 500058, India.
E-mail addresses: bhaktvidya@gmail.com (B. Mishra), rajdeep.sarkar@gmail.com (R. Sarkar), vajindersingh83@gmail.com (V. Singh), deepakmet@yahoo.com (D. Kumar), amritendu08@gmail.com (A. Mukhopadhyay), madhu.vemuri@dmrl.drdo.i (V. Madhu), mjnvprasad@iitb.ac.in (M.J.N.V. Prasad).

https://doi.org/10.1016/j.msea.2022.144324
Received 10 June 2022; Received in revised form 6 November 2022; Accepted 8 November 2022
Available online 12 November 2022
0921-5093/© 2022 Elsevier B.V. All rights reserved.

in fully recrystallized alloys by subsequent ageing at 723 K–923 K [11–14]. However, a prolonged ageing treatment reduces ductility of LDS alloys due to precipitation of intergranular κ-carbides [15,16]. It is also reported that concurrent κ-carbides and ferrite precipitation affects the recrystallization kinetics. This phenomenon has been exploited by several researchers to control the primary recrystallization of austenite in Ni-free LDS for achieving better strength-ductility combination [17–20]. Further, Ni addition in Fe–Mn–Al–C based LDS facilitates precipitation of finer Ni-rich ordered B2 particles in the temperature range of 973 K–1173 K [7–10]. It seems to be that Ni addition provides an opportunity to further exploit thermo-mechanical processing window to achieve better properties by fine tuning the recrystallization process after cold deformation, which is not yet explored.

Zhang et al. [17] utilized concurrent second phase precipitation and controlled recrystallization in Fe–26Mn-10.6Al-1.2C alloy to attain better strength-ductility combination. A processing window was explored by varying deformation (rolling reduction at room temperature) and subsequently, the annealing temperature and time. To optimize the recrystallization kinetics thickness reduction levels of 60% and annealing temperature window of 1123 K–1173 K were explored. The best strength-ductility combination was reported for annealing temperature and time combination of 1143 K and 30 s, respectively [17]. On similar lines, synergistic effect of partial recrystallization and κ-carbide strengthening resulting into higher yield strength and sufficient ductility in Ni-free LDS was reported by annealing for shorter duration of time (3 min) at 1123 K–1223 K [18]. Brasche et al. [19] reported change in morphology and precipitation of κ-carbides in deformed and recrystallized austenite regions. Also, their interplay with recrystallization was utilized to attain higher strength-ductility combination in Ni-free LDS. It was also reported that Ni containing LDS shows fine ordered B2 precipitation during thermo-mechanical processing [7–10]. Similar to κ-carbide, concurrent B2 precipitation can be exploited for obtaining superior properties by controlling the recrystallization process post cold rolling deformation, which is not systematically addressed in literature.

In view of the gaps in literature, as mentioned above, the present work is aimed to explore the effect of Ni addition *vis-à-vis* thermo-mechanical processing on microstructure evolution and their effects on mechanical properties include hardness and tensile response. A comparison study has been made with Ni-free LDS. In this context, three different compositions Fe–28Mn–9Al-0.9C, Fe–28Mn–9Al-0.9C–5Ni and Fe–15Mn–9Al-0.9C–15Mn (all elements are in wt. %) namely, 28Mn0Ni, 28Mn5Ni and 15Mn5Ni, respectively were investigated to understand the effects of Mn and Ni on secondary phase precipitation and recrystallization of austenite and subsequently on mechanical behavior of LDS alloys.

2. Materials and experimental methods

2.1. Thermo-mechanical processing

For the present study, three low-density steel grades with different Mn and Ni content were prepared by vacuum induction melting route. The chemical composition of the three alloys determined by wet chemical analysis and emission spectroscopy is provided in Table 1. The sound portion of the 30 kg ingot was homogenized at 1423 K for 2.5 h and then hot rolled at 1323 K. This processing condition is hereafter referred to as HR. The final thickness of hot rolled plate was about 13 mm. The details of the hot rolling process were reported elsewhere [21]. Samples were cut from the hot-rolled plates and were subsequently cold rolled to ~50% reduction in thickness (~6 mm thick sheets) by subjecting to several passes. It is important to note that 15Mn5Ni alloy has experienced cracking beyond 50% cold reduction; hence, further reduction was not carried out on any of the chosen alloys for comparison reasons. The samples after cold rolling are denoted with HR + CR. Subsequently, the cold rolled sheets were heat treated in the temperature range of 873 K–1273 K for 1 h followed by water quenching. The heat-treated samples are henceforth referred with the heat treatment temperature.

Table 1
The chemical composition of the three different LDS alloys used for the present study.

Alloy designation	Elements in weight %				
	Mn	Al	C	Ni	Fe
28Mn0Ni	28.3 ± 0.4	9.0 ± 0.05	0.9 ± 0.01	–	Balance
28Mn5Ni	28.1 ± 0.4	8.6 ± 0.04	0.9 ± 0.01	4.8 ± 0.04	Balance
15Mn5Ni	14.7 ± 0.3	9.8 ± 0.05	0.9 ± 0.01	4.1 ± 0.05	Balance

2.2. Microstructure characterization

Orientation imaging microscopy (OIM) was performed on cold rolled and heat-treated samples using Zeiss Supra 55 FEG-Scanning electron microscope (SEM) equipped with EDS and Nordley EBSD detectors from Oxford. Samples for OIM were polished to colloidal silica of ~100 nm surface finish. A step size of 120 nm and 250 nm was employed on an area of ~100 × 100 μm^2 and ~200 × 200 μm^2 for fine- and coarse-grained samples, respectively. Three scans for each sample were used to analyze the microstructure. The collated data were then analyzed using TSL-OIM™ software for grain orientations, grain size and twin fractions. Detailed analysis includes kernel average misorientation (KAM) representing strain localization and grain orientation spread (GOS) of 2° criterion to delineate the recrystallized microstructure. Grain size of the recrystallized grains was determined using linear intercept method. To calculate the volume fraction of twins, coherent site lattice (CSL) Σ3 boundaries were selected. Based on the fraction of Σ3 boundaries, twin fraction was reported. In addition, X-ray diffraction studies were carried out on selected samples using Co-K_α radiation at 45 kV and 135 mA with a beam size of ~2 mm in a Rigaku XRD instrument.

Transmission electron microscopy (TEM) studies were carried out to understand and identify the phases present in the heat-treated samples. The specimens for TEM examination were prepared by twin jet electropolishing technique using mixed acid solution as electrolyte at 243 K (−30 °C). A FEI TECNAI 20T G2 transmission electron microscope operating at 200 kV equipped with EDAX energy dispersive X-ray spectroscopy was used for TEM study. Post tensile fractography studies were performed on selected fractured specimens of the alloys using Zeiss EVO 18 SEM.

2.3. Mechanical characterization

Hardness and tensile tests of heat-treated samples were conducted to evaluate the change in mechanical properties for the chosen alloys upon cold working and heat treatment. The bulk hardness was measured as per ASTM E−140-02 using an AFFRI Vickers hardness tester under a peak load of 30 kg (~300 N) for a dwell time of 15 s. Hardness values averaged over ten measurements on each sample were reported. Flat tensile specimens with a gauge length of 30 mm, width 6 mm and thickness 4–5 mm were prepared and tested as per ASTM E8M standard for evaluating strength and ductility of the chosen alloys in different processing conditions. Tensile tests were conducted at room temperature under a constant cross-head speed of 0.5 mm min^{-1} (initial strain rate of ~3×10^{-4} s^{-1}) using Walter + Bai^{AG} testing machine. The tensile tests were triplicated for each condition and the average of three readings was reported. Additionally, interrupted tensile tests were carried out on selected specimens to understand the role of microstructure in general and second phase in particular on evolution of microstructure with progress of deformation.

3. Results

3.1. Microstructural evolution

3.1.1. Effect of cold reduction

Microstructures of all the three alloys under investigation in hot rolled (HR) condition were already reported elsewhere [21]. Summary of the findings include: (i) all the three alloys exhibited equiaxed and similar grain sizes of γ-matrix (d ~5–7 μm) with nearly random texture in HR condition; (ii) Ni addition led to formation of considerable amounts of B2 phase in the form of elongated stringers (bamboo-like morphology) and polygonal particles in γ-matrix, where15Mn5Ni alloy contained higher B2 fraction (~12%) than that of in 28Mn5Ni (~4%), (iii) similarly, 15Mn5Ni alloy possessed higher volume fraction of κ-carbides followed by 28Mn5Ni and 28Mn0Ni alloys, and (iv) the microstructures of higher Mn LDS alloys consisted of higher fraction of twins (~28–40%) than in 15Mn5Ni alloy.

Fig. 1 presents the OIM-EBSD results of the three alloys in HR + CR condition. Fig. 1a–c, d-f and g-i show the colored inverse pole figure (IPF), image quality (IQ) and phase contrast maps, respectively of 28Mn0Ni, 28Mn5Ni and 15Mn5Ni alloys. As noted earlier in HR condition, 28Mn0Ni alloy is a single-phase austenite whereas both 28Mn5Ni and 15Mn5Ni alloys exhibit two phase (γ + B2) microstructure with higher fraction of B2 in the latter alloy in HR + CR condition too. Irrespective of the alloy content, the austenite grains deformed mostly by planar slip as reflected by parallel slip bands marked by an arrow in Fig. 1b. There were no traces of deformation twinning and martensite transformation in any of the cold rolled alloys and is consistent with the earlier reports on similar grades of LDS [22,23]. However, cold rolling led to substantial changes in the morphology of B2 stringer seen in 15Mn5Ni. It is noticed that the B2 stringer underwent concurrent deformation with austenite grains, which resulted in significant reduction in the average thickness of B2 stringer from ~8 μm to ~2 μm and eventually sheared and fragmented at higher strain. The inset in Fig. 1g shows the OIM-IPF map for a selected B2 phase (region marked by a rectangular box) alone in cold rolled 15Mn5Ni alloy. One of such typical stringers having near <100> orientation undergoes shearing and subsequent fragmentation (~35–45° to rolling direction) at two different locations marked by arrows in the inset. This clearly indicates that B2 being an ordered intermetallic phase yet shares considerable load during deformation and can undergo progressive microstructural changes ranging from dislocation accumulation, substructure development to fragmentation of stringers during loading.

3.1.2. Effect of heat treatment

Fig. 2a and b shows the XRD patterns of 28Mn0Ni and 15Mn5Ni alloys, respectively in both cold rolled (HR + CR) and heat-treated conditions. Both 28Mn0Ni and 15Mn5Ni alloy samples in cold rolled condition showed the peaks at Bragg angles corresponding to γ-phase and γ+B2 phases, respectively. However, upon heat treatment at 973 K additional peaks corresponding to κ-carbides and B2 appeared in both the alloys, and this observation is consistent with the earlier reports on similar steel grades [18,24–26]. On further heat treatment at 1173 K, the XRD pattern of 28Mn0Ni alloy showed peaks corresponding to γ-phase only, whereas 15Mn5Ni alloy exhibited XRD peaks corresponding to both γ and B2 phases, but no noticeable peaks corresponding to κ-carbides. Surprisingly, the diffraction peaks corresponding to B2 were more intense in 15Mn5Ni alloy sample heat treated at 973 K than in the same alloy heat treated at 1173 K. This indicates that higher volume fraction of the secondary phases at lower annealing temperature.

Microstructure evolution in all the three alloys under various heat treatment conditions was examined in-detail and the results are presented in Figs. 3–7. Three main characteristic microstructural features were observed in the heat-treated samples, namely (a) deformed austenite, (b) recrystallized austenite along with annealing twins and (c) secondary B2 phase. Two different types of B2 precipitation were noticed, namely fine intragranular B2 (size: 50–200 nm) and coarser intergranular B2 (size: 0.5–1.5 μm). The former ones could not be indexed with SEM-EBSD characterization. In addition, B2 stringers were present prior to heat treatment in Ni containing low-density steels.

Fig. 3 shows the OIM-EBSD results of the three alloys in HR + CR+1023 K condition. Fig. 3a–c, d-f and g-i show the colored IPF, phase maps and kernel average misorientation (KAM) maps, respectively of 28Mn0Ni, 28Mn5Ni and 15Mn5Ni alloys. In Fig. 4, for brevity reasons, just the colored IPF maps of the three alloys in HR + CR+973 K (Fig. 4a, c,e) and HR + CR+1173 K (Fig. 4b,d,f) conditions are presented. To

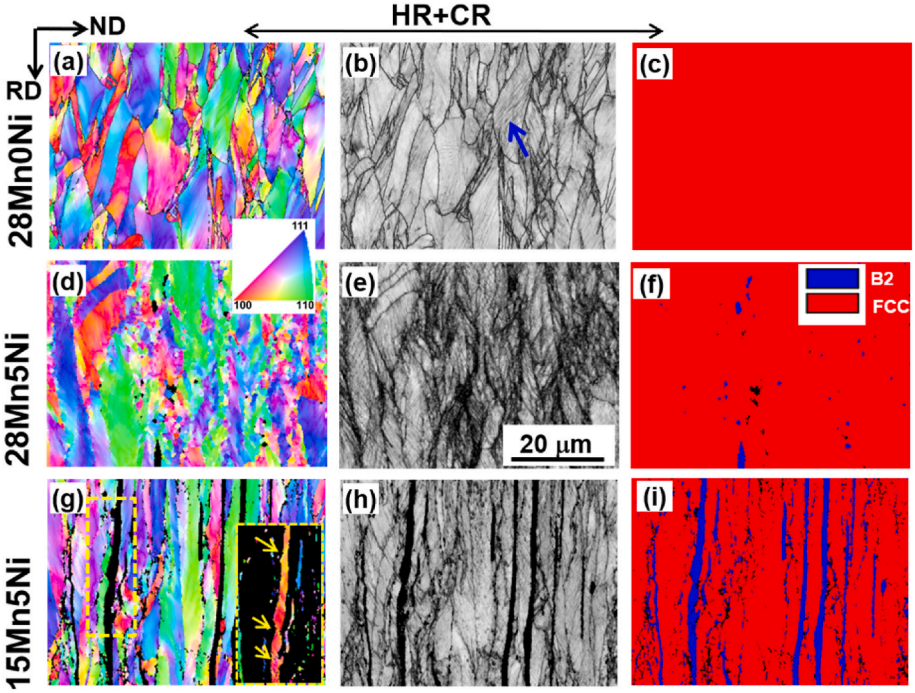


Fig. 1. Colored inverse pole figure (IPF) maps (for austenite phase), image quality (IQ) maps and phase contrast maps, respectively of (a–c) 28Mn0Ni, (d–f) 28Mn5Ni and (g–i) 15Mn5Ni alloys in HR + CR condition. Inset of (g) shows the IPF map of B2 phase alone. The arrows in yellow color shown in the inset of (g) indicate shearing and fragmentation of B2 phase. Whereas, the arrow in blue color shown in (b) reveal the slip bands. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

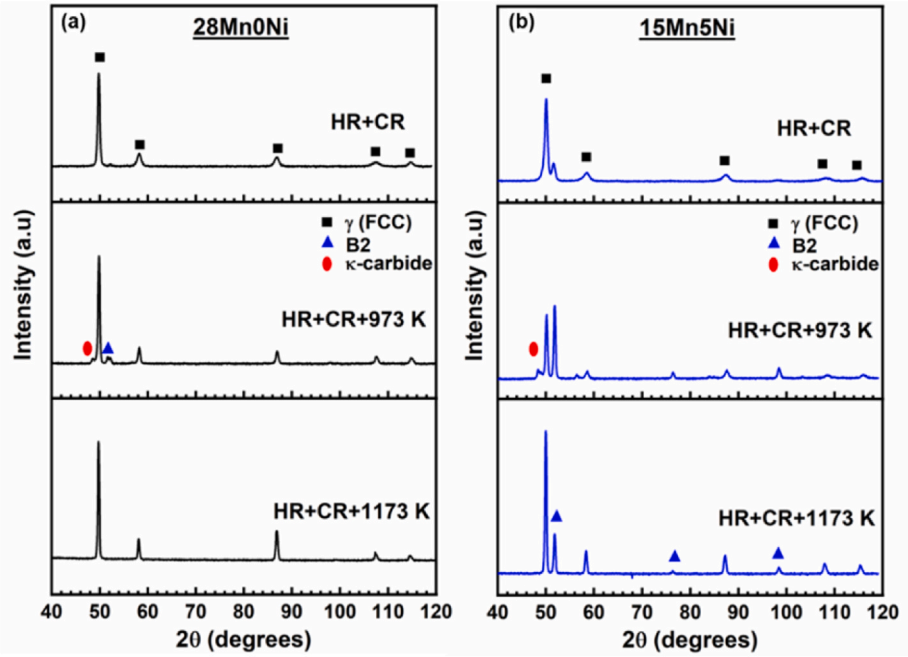


Fig. 2. X-ray diffraction patterns of (a) 28Mn0Ni and (b) 15Mn5Ni alloy in HR + CR as well as in selected heat-treated conditions.

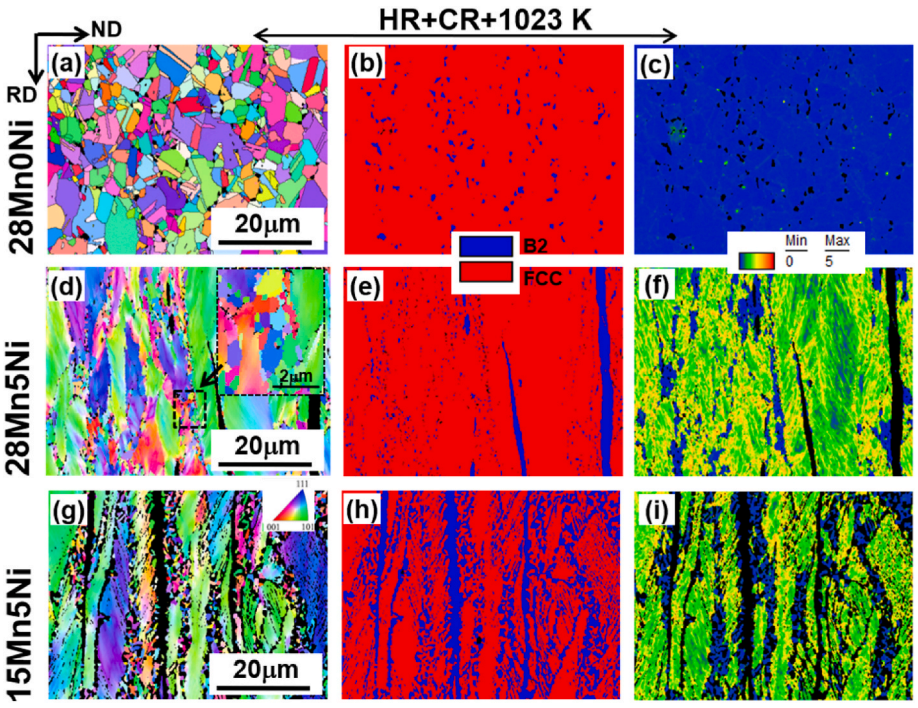


Fig. 3. Colored IPF (for austenite phase), phase contrast and KAM maps of (a–c) 28Mn0Ni, (d–f) 28Mn5Ni and (g–i) 15Mn5Ni alloys, respectively in HR + CR+1023 K condition. Inset of (d) shows the colored IPF map at higher magnification revealing the fine recrystallized austenite grains.

highlight the dissolution of B2 particles with increase in heat treatment temperature, the phase maps of 15Mn5Ni alloy in HR + CR+973 K and HR + CR+1173 K conditions are shown in Fig. 4g and h, respectively. Dissolution rate of finer B2 particles was seen to be higher than that of B2 stringers. The recrystallization fraction was calculated based on KAM maps as well as GOS <2° criteria. Fig. 5 illustrates the evolution of microstructural features (quantified using the OIM-EBSD data) such as recrystallization fraction of austenite grains (Fig. 5a), the mean grain size of recrystallized austenite grains (Fig. 5b), twin fraction (Fig. 5c) and secondary phase (B2) volume fraction (Fig. 5d) as a function of heat treatment temperature and alloy composition. The volume fraction of prior B2 phase in both 28Mn5Ni and 15Mn5Ni are reported with horizontal broken lines in Fig. 5d.

The following points were noted on the effect of subsequent heat treatment on microstructural evolution in the cold rolled three alloys: (i) The degree of recrystallization of austenite grain structure in a given alloy increased with the heat treatment temperature; for example, the cold rolled 28Mn0Ni alloy microstructure transformed from partial

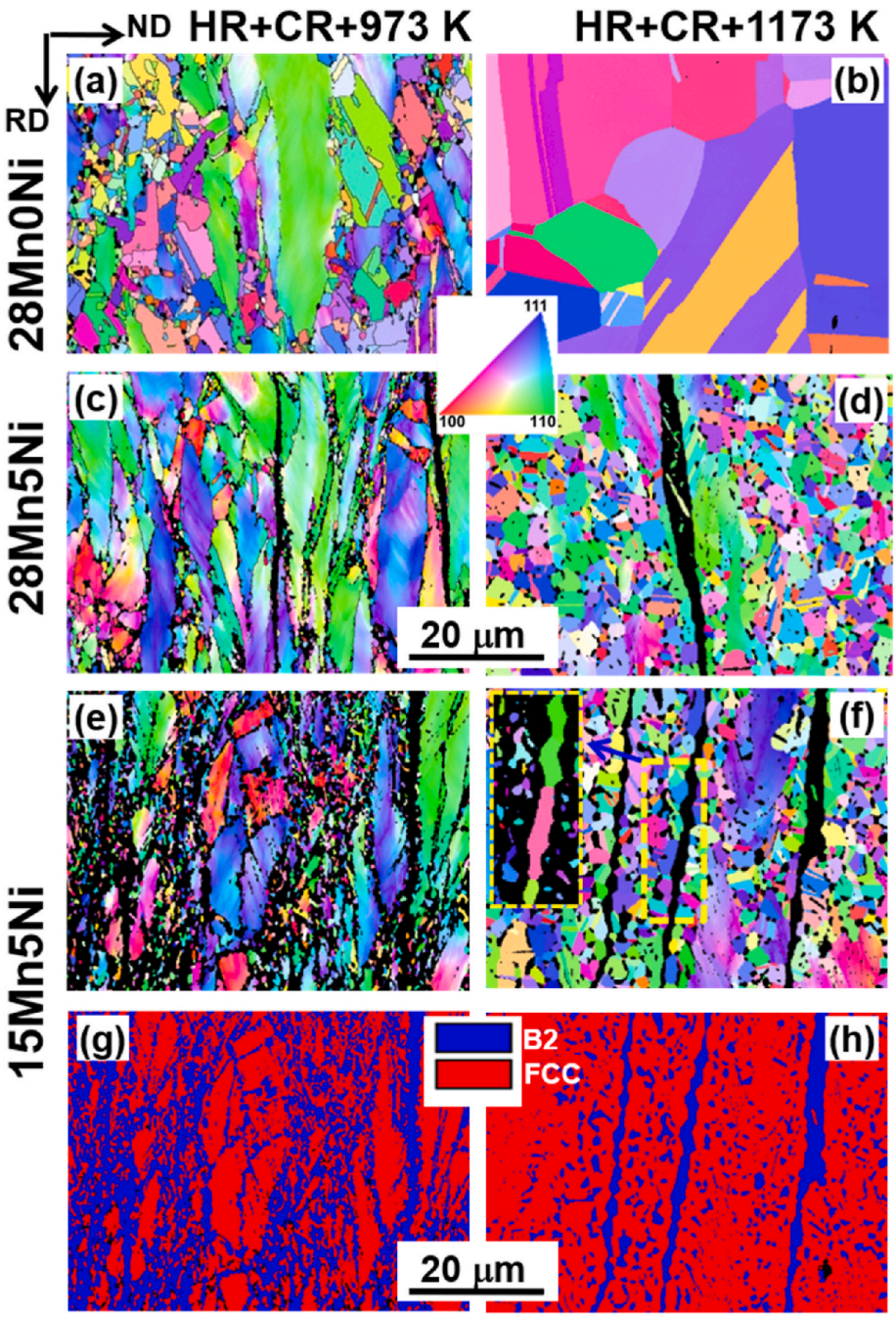


Fig. 4. Colored IPF maps (for austenite phase) of (a,b) 28Mn0Ni, (c,d) 28Mn5Ni and (e,f) 15Mn5Ni alloys in HR + CR+973 K and HR + CR+1173 K conditions, respectively. (g,h) phase maps of 15Mn5Ni alloy in HR + CR+973 K and HR + CR+1173 K conditions, respectively. Inset of (f) shows the colored IPF map for B2 phase alone revealing an extended recovery of its grains.

recrystallization at 973 K to complete recrystallization at 1023 K followed by significant grain growth at 1173 K, which indicates an increased mean grain size of the recrystallized austenite with temperature. (ii) Surprisingly, the volume fraction of recrystallized austenite grains at a given temperature was noted to be higher in nearly B2-free alloy than that of in B2-containing alloys, i.e., 28Mn0Ni alloy exhibited extensive recrystallization as compared to in 28Mn5Ni and 15Mn5Ni alloys. (iii) The volume fraction of annealing twins increased with the heat treatment temperature and was found to be highest in 28Mn0Ni alloy and least in 15Mn5Ni alloy at any given temperature. (iv) Significant amounts of secondary B2 phase precipitated out in all three alloys, even in 28Mn0Ni alloy upon heat treatment at 973 K; however, its volume fraction decreased gradually with further heat treatment at higher temperatures. This observation corroborates the XRD results (Fig. 2). Though Ni addition facilitates precipitation of B2 phase, it was more pronounced at lower Mn levels i.e., in 15Mn5Ni alloy during heat treatment at 973 K. It is also interesting to notice prominent coarser precipitate (B2) rich zone around deformed austenite in 15Mn5Ni alloy, whereas in 28Mn5Ni alloy fine secondary B2 was precipitated mostly along grain boundaries. In addition, the inset in Fig. 4f shows the colored IPF map for B2 phase alone, and it illustrates the presence of both prior B2 phase and new B2 precipitates formed upon heat treatment at 1173 K. The prior B2 phase grains, which deformed heavily along with γ-matrix in cold rolling, were fully recovered and developed bamboo-type grain morphology upon heat treatment. Interestingly, the new B2 precipitates showed single crystal type orientations, but displayed some

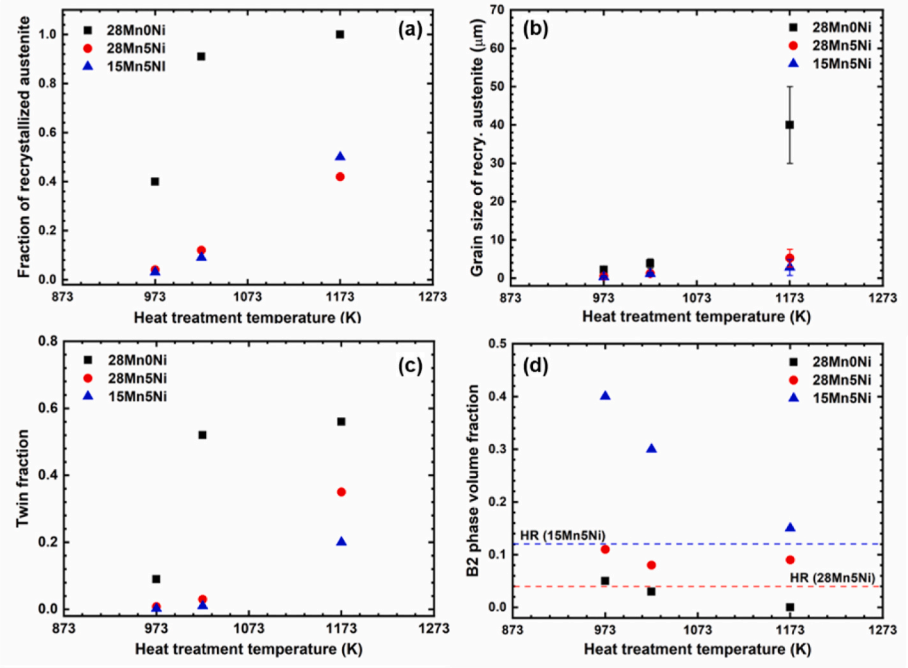


Fig. 5. The variation in (a) recrystallization fraction of austenite, (b) grain size of recrystallized austenite, (c) twin fraction and (d) secondary B2 phase fraction in 28Mn0Ni, 28Mn5Ni and 15Mn5Ni alloys as a function of heat treatment temperature.

orientation relationship with the adjacent γ-grains.

In order to characterize the deformed austenite regions and finer microstructural features such as κ-carbides and B2 particles which were not revealed at SEM-EBSD scale, TEM studies were carried out on selected heat-treated samples. Fig. 6a–d, e-f and g-h show the TEM images of 28Mn0Ni, 28Mn5Ni and 15Mn5Ni alloy samples heat treated at 973 K. Fig. 6b,d,f,h are dark-field TEM micrographs and the rest are bright-field images. The dark-field images were captured from (010) satellite spots corresponding to κ-carbide in selected area electron diffraction (SAED) patterns shown as insets in Fig. 6a–d. As noticed in EBSD examination (Fig. 4), there were mixed regions of recrystallized and deformed austenite grains in all the alloy samples heat treated at 973 K. In recrystallized austenite regions, fine uniformly distributed intragranular κ-carbides were seen (shown by dotted (orange) arrow in Fig. 6b), while the deformed austenite region contained inhomogeneous and relatively coarse κ-carbides of size 10–60 nm (Fig. 6d). The size of κ-carbides in deformed austenite regions was found to be higher in 15Mn5Ni alloy than that compared to in 28Mn0Ni and 28Mn5Ni alloys under similar conditions. The observations of coarser and higher volume fraction of κ-carbides in deformed austenite and finer ones in recrystallized region are consistent with earlier studies shown in Ni-free LDS [17–19]. In addition to κ-carbides, intergranular B2 particles were observed in 28Mn0Ni alloy samples heat treated at 973 K. One such region marking presence of intergranular B2 particles identified by solid (white) arrow is shown in Fig. 6a. Similarly, fine intragranular B2 (5–20 nm sizes) marked by broken (yellow) arrow, intergranular B2 (50–200 nm sizes) particles as well as κ-carbides were noticed in both 28Mn5Ni (Fig. 6e and f) and 15Mn5Ni (Fig. 6g and h) alloy samples heat-treated at 973 K. The microstructure evolution in all the three alloys under heat treatment at higher temperature of 1173 K is shown in Fig. 7. It is noted that no secondary phase precipitates were seen in 28Mn0Ni alloy (Fig. 7a and b). In case of Ni containing steels heat-treated at 1173 K, fraction of intragranular B2 precipitation decreased with increase in recrystallized austenite fraction (Fig. 7c–f). This indicates that highly strained regions are preferential nucleation sites for these precipitates. Furthermore, the composition profile of B2 was taken using TEM-EDS and is provided in Table 2. This elemental analysis indicated B2 as FeAl based phase but contains significant amounts of Mn and Ni in it, whose content varied with their contents in the steel. Furthermore, the B2 phase enriched with Ni content with increasing the heat treatment temperature.

3.2. Mechanical behavior

3.2.1. Hardness variation

Fig. 8a shows the variation in bulk hardness on Vickers scale (~300 N peak load) as a function of degree of cold work under rolling at room temperature. The initial hardness values of 28Mn0Ni, 28Mn5Ni and 15Mn5Ni alloys in HR condition were noted to be ~256 ± 5 VHN, ~268 ± 5 VHN and ~375 ± 6 VHN respectively [21]. With progress of deformation under cold rolling the hardness of all the three alloys increased and surprisingly, saturated to around 450 VHN at 50% of cold reduction. At 50% cold reduction, the hardness of 15Mn5Ni alloy was ~20 VHN higher than that of 28Mn0Ni alloy. It is also interesting to note that in the initial cold reduction regime the rate of work hardening in Ni free alloy (28Mn0Ni) was lower as compared to in the Ni containing alloys (28Mn5Ni and 15Mn5Ni). However, later the rate of work hardening increased rapidly in 28Mn0Ni alloy with further cold reduction.

Fig. 8b illustrates the effect of subsequent heat treatment temperature on evolution of hardness in all the three alloys subjected to 50% cold reduction. The significant findings are as follows: (i) Upon subsequent heat treatment at 873 K, the hardness of Ni-containing alloys (28Mn5Ni and 15Mn5Ni) was increased to 500 VHN, while there was no noticeable change in hardness of 28Mn0Ni alloy. (ii) Under a given heat treatment, hardness of Ni containing alloys was significantly higher than that of 28Mn0Ni alloy, (iii) both the Ni containing alloys showed almost similar trend in drop in hardness with the heat treatment temperature, and (iv) the drop-in hardness with temperature was more prominent in Ni-free (28Mn0Ni) alloy. The variation in hardness of LDS alloys as a function of heat treatment temperature could be related to substantial and differential microstructural changes, as noted in previous section, such as precipitation and dissolution of secondary phases (κ-carbides and B2 particles), recrystallization and grain growth phenomena etc., occurring in these alloys depending on the composition of steel and is

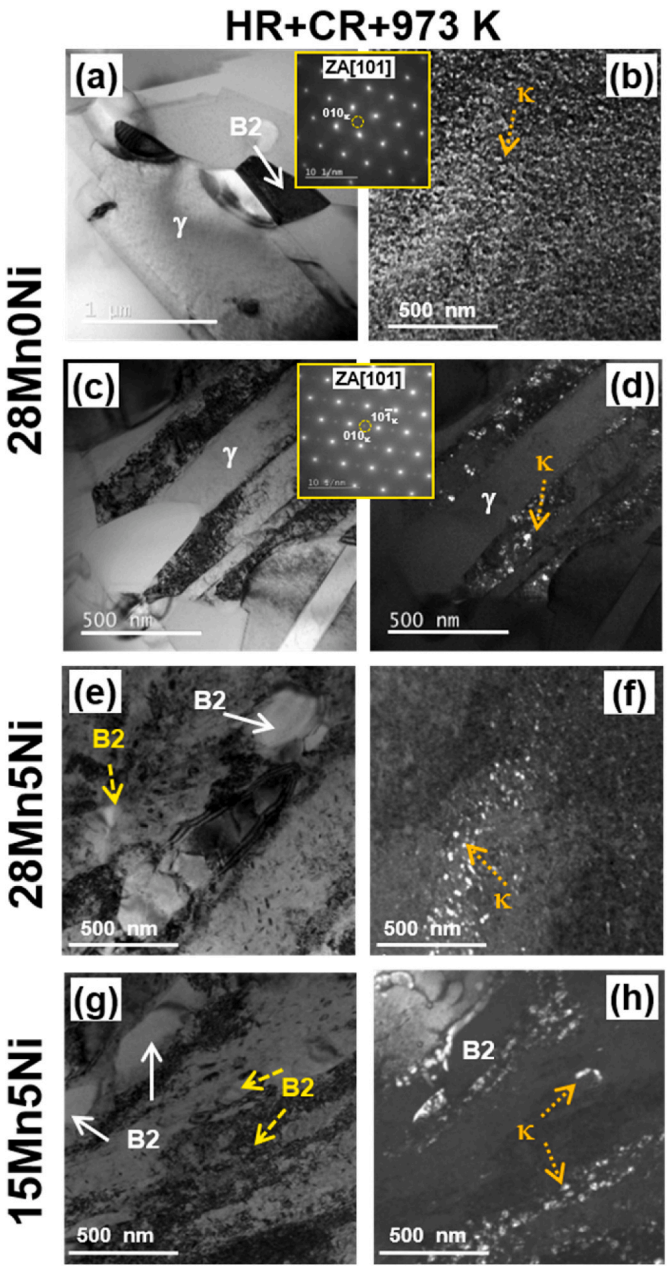


Fig. 6. TEM images of (a–d) 28Mn0Ni, (e,f) 28Mn5Ni and (g,h) 15Mn5Ni alloys in HR + CR+973 K condition. Here, (b,d,f,h) are dark-field images and the rest are bright-field images. Insets of (a–d) show selected area diffraction pattern (SADP) taken from the austenite grains. Dark-field images were captured from $(010)_{\kappa}$ satellite spot as shown in SADP. Intergranular and intragranular B2 particles are marked with broken (yellow) arrows and solid (white) arrows, respectively, and κ-carbides are identified by dotted (orange) arrows. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

discussed in-detail later.

3.2.2. Tensile response

Fig. 9a and b shows engineering stress-engineering strain curves of all the three alloys in HR + CR (50% cold reduction) and in selected heat treatment conditions, respectively. The major findings include: (i) All the three alloys in HR + CR condition exhibited higher yield stress (0.2% offset) in the range of 1.35–1.67 GPa and tensile strengths of 1.5–1.8 GPa but limited tensile elongations of 6–11% with negligible strain hardening. Among the three alloys, 15Mn5Ni alloy displayed relatively

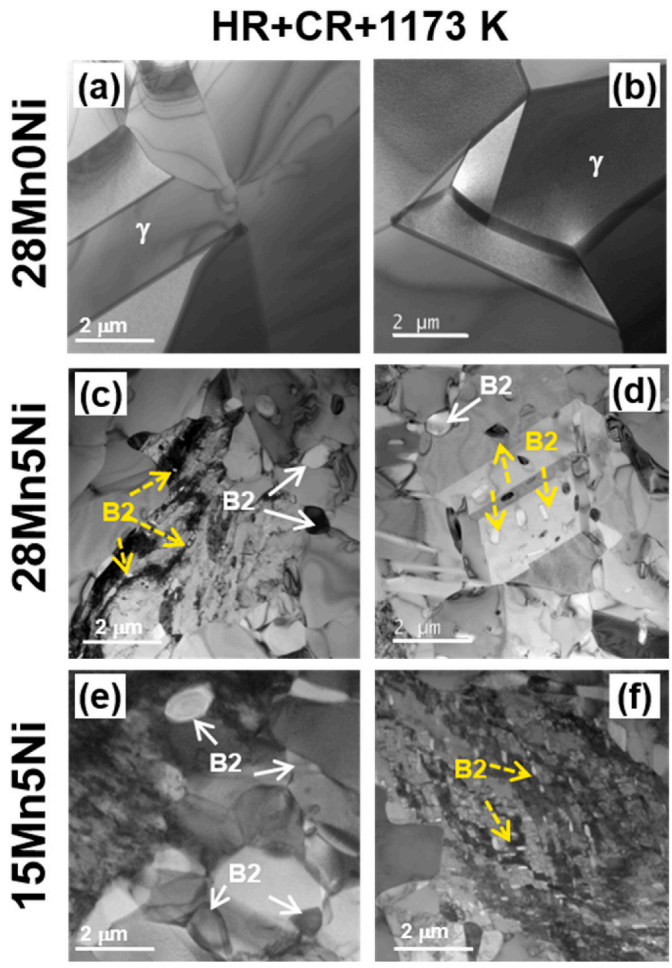


Fig. 7. TEM bright-field images captured at two different locations for (a,b) 28Mn0Ni, (c,d) 28Mn5Ni and (e,f) 15Mn5Ni alloys in HR + CR+1173 K condition.

Table 2
The elemental composition analysis obtained by TEM-EDS for B2 particles in two LDS alloys in two different heat-treated conditions. (Note: The balance is Fe content).

Alloy	Heat treatment temperature	Composition of B2 (in wt. %)	Composition of B2 (in at. %)
28Mn5Ni	973 K	Mn:16.3	Mn: 14.7
		Al: 12.7	Al: 23.3
		Ni: 24.6	Ni: 20.8
	1173 K (Particle-1)	Mn:16.5	Mn: 14.9
		Al: 14.2	Al: 25.7
		Ni: 26.1	Ni: 21.8
	1173 K (Particle-2)	Mn:15.4	Mn: 13.7
		Al: 14.8	Al: 26.7
		Ni: 27.5	Ni: 22.8
15Mn5Ni	1173 K (Particle-1)	Mn: 10.5	Mn: 9.5
		Al:12.8	Al: 23.5
		Ni: 20.2	Ni: 17.0
	1173 K (Particle-2)	Mn: 8.7	Mn: 7.8
		Al:13.5	Al: 24.6
		Ni: 22.0	Ni: 18.5

higher strengths and lower ductility. It was noted that the uniform strain (i.e., just before necking) values in HR + CR condition were quite low and were less than 3% in all the alloys. (ii) 28Mn0Ni alloy showed lower strength, higher ductility and high strain hardening ability in all heat-treated conditions. The maximum tensile elongation to failure of ~86% was attained by this alloy in fully recrystallized and grain growth

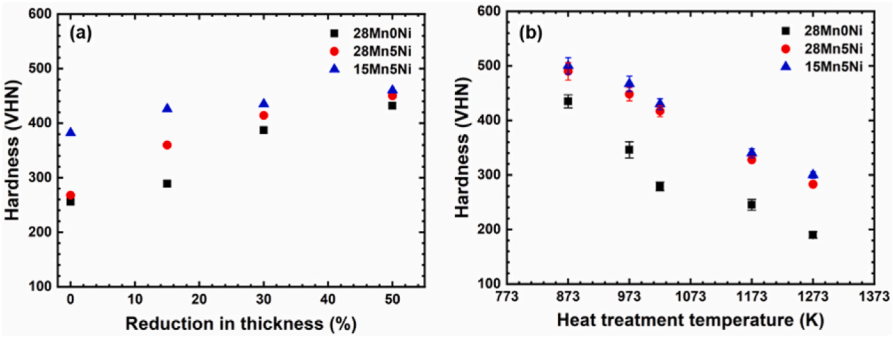


Fig. 8. The variation in hardness in 28Mn0Ni, 28Mn5Ni and 15Mn5Ni alloys as a function of (a) degree of cold rolling (i.e., %reduction in thickness) and (b) heat treatment temperature.

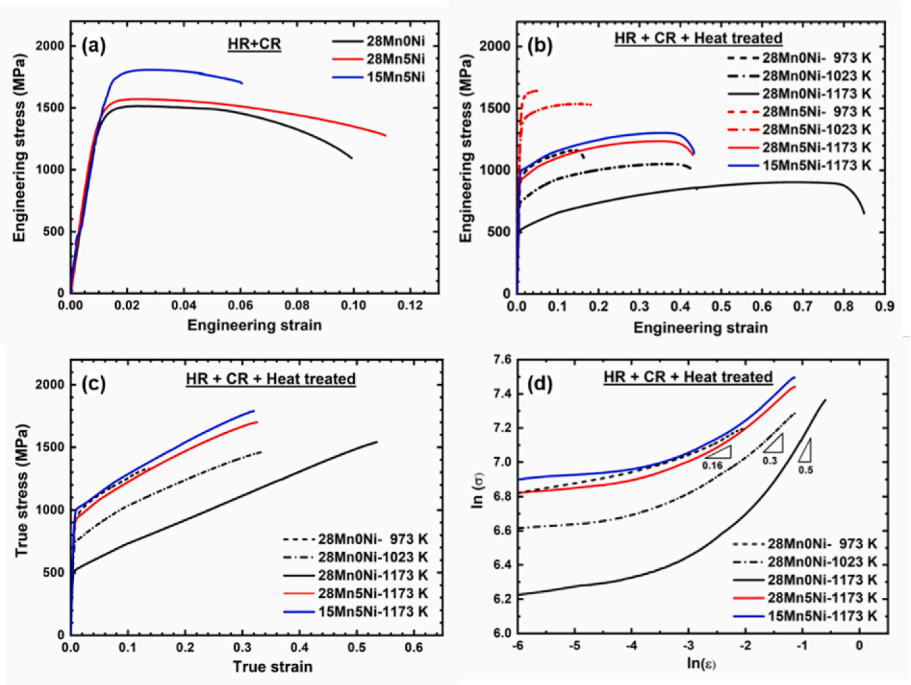


Fig. 9. Engineering stress-engineering strain curves of 28Mn0Ni, 28Mn5Ni and 15Mn5Ni alloys in (a) HR + CR condition and (b) heat treated condition. (c) True stress-true strain curves and (d) log(σ) versus log(ε) plot for the three alloys in selected heat treated conditions.

condition (HR + CR+1173 K). (iii) Surprisingly, upon heat treatment to 973 K and 1023 K the 15Mn5Ni alloy behaved like brittle material with tensile elongation <1% and hence, the stress-strain curves of this alloy in these heat-treated conditions could not be recorded. Whereas, the 28Mn5Ni alloy in same heat-treated conditions showed significant ductility. (iv) Although both 28Mn0Ni and 28Mn5Ni alloys in HR + CR condition exhibited similar strength levels and tensile elongations, interestingly, the latter alloy exhibited much higher strengths than that of 28Mn0Ni alloy in all heat-treated conditions, and (v) Upon heat treatment to higher temperature of 1173 K, the 15Mn5Ni alloy regained its ductility and showed comparable stress-strain behavior as that of 28Mn5Ni alloy. The yield stress (YS), the tensile strength (TS), total tensile elongation to failure (TE) derived from the stress-strain curves of the three alloys in various conditions are listed in Table 3.

The effects of (i) heat treatment temperature on strain hardening capability of 28Mn0Ni alloy and (ii) the alloy composition on strain hardening capability of all the three alloys in fully recrystallized condition i.e., heat treated to a fixed temperature of 1173 K were analyzed separately using true stress-true strain results obtained from standard conversion equations valid up to the onset of necking, and the curves are

Table 3
Tensile properties of 28Mn0Ni, 28Mn5Ni and 15Mn5Ni alloys in both cold rolled and subsequent heat-treated conditions.

Condition	Alloy	YS (MPa)	UTS (MPa)	TE (%)	n
HR + CR	28Mn0Ni	1350 ± 10	1500 ± 10	10 ± 0.4	
	28Mn5Ni	1394 ± 5	1571 ± 5	11 ± 0.4	
	15Mn5Ni	1665 ± 12	1800 ± 15	7 ± 0.5	
HR + CR+973 K	28Mn0Ni	884 ± 6	1162 ± 10	23 ± 0.5	0.16
	28Mn5Ni	1450 ± 15	1645 ± 10	5 ± 1	
HR + CR+1023 K	28Mn0Ni	732 ± 2	1051 ± 5	36 ± 2	0.30
	28Mn5Ni	1289 ± 2	1537 ± 2	17 ± 1	
HR + CR+1173 K	28Mn0Ni	504 ± 5	905 ± 8	86 ± 2	0.50
	28Mn5Ni	900 ± 4	1236 ± 5	39 ± 1	0.30
	15Mn5Ni	971 ± 5	1303 ± 12	36 ± 1	0.30

Note: Tensile properties of 15Mn5Ni alloy specimens heat treated at 973 K and 1023 K are not available, as this alloy exhibited brittle nature under these conditions. Where, YS: yield stress at 0.2% offset, UTS: ultimate tensile strength, TE: total tensile elongation to failure and n: strain hardening exponent.

shown in Fig. 9c. In addition, the effects of heat treatment and composition on strain hardening ability of LDS were evaluated from plastic deformation region of their respective true stress-true strain data in the form of ln(σ) *versus* ln(ε), and compared in Fig. 9d. All the tensile flow curves showed typical two slope characteristics with strain hardening exponent '*n*' value quite small (<0.1) at low strain regime. However, the *n* value at larger strain was found to be significant and varied with the composition of the steel and the heat treatment temperature. For example, in case of 28Mn0Ni alloy the *n* value increased from 0.16 to 0.50 with increasing the heat treatment temperature from 973 K to 1173 K (Fig. 9c). Both the Ni-containing alloys (28Mn5Ni and 15Mn5Ni) exhibited nearly identical but relatively lower *n* value of ~0.30 as compared to the 28Mn0Ni alloy heat treated to same temperature of 1173 K (Fig. 9d). The values of *n* are reported in Table 3. The analysis indicates higher strain hardening ability of Ni-free (28Mn0Ni) alloy than that of Ni-containing LDS alloys. This observation is consistent with the earlier reports [27–29]. The differences in strain hardening capability could primarily be related to the changes in microstructural constituents such as the recrystallization fraction of austenite, the grain size of the recrystallized grains, κ-carbides and B2 particles in these alloys and their effect on dislocation substructure evolution during deformation and is discussed in-detail later.

3.3. Post tensile deformed microstructures

As mentioned in previous section, 15Mn5Ni alloy specimens heat treated at 973 K and 1073 K displayed negligible ductility (<1%), whereas 28Mn0Ni and 28Mn5Ni alloy specimens heat treated to the same conditions showed remarkable tensile elongations. Hence, fractography of all the three alloy tensile specimens heat treated at 1023 K but failed in tension at room temperature is presented in Fig. 10a–c, respectively. The morphology of fracture surface of 15Mn5Ni alloy specimen (Fig. 10c) was distinctive from that of the other two alloy specimens (Fig. 10a and b) and is in accordance with the tensile response of the alloys. Both 28Mn0Ni and 28Mn5Ni alloys exhibited dimple morphology whereas quasi-cleavage fracture was seen in 15Mn5Ni alloy specimen. This indicates that the failure in both 28Mn0Ni and 28Mn5Ni alloys was ductile in nature. Fracture in 15Mn5Ni alloy was seemed to crack initiation-controlled with fast propagation of cracks. Large numbers of microcracks were seen on fracture surface of 15Mn5Ni alloy specimen. It is to be noted that 15Mn5Ni alloy showed much lesser recrystallization of austenite matrix and large volume fraction of secondary phase precipitation along grain boundaries forming precipitate rich zones. Further, cross section of the failed tensile specimen of 15Mn5Ni alloy heat treated at 1023 K was examined in-detail. Fig. 10d shows a scanning electron micrograph of one such crack regions revealing presence of intergranular cracks. Fig. 10e and f shows the OIM-IPF and phase maps, respectively of another cracked region, clearly indicating propagation of crack along γ/B2 interphase boundaries.

In addition, microstructures of selected interrupted tensile specimens of 28Mn0Ni and 15Mn5Ni alloys were examined in order to understand the dislocation substructure evolution. Figs. 11 and 12 show the post-tensile deformed microstructures of 28Mn0Ni and 15Mn5Ni alloys, respectively. Fig. 11a shows TEM bright-field image of 28Mn0Ni-HR + CR+973 K sample deformed up to a true strain of ε~0.14. Fig. 11b–e and 12a-d show the TEM bright-field images, respectively of 28Mn0Ni and 15Mn5Ni alloys heat treated at 1173 K, as a function of the applied strain. There was progressive substructure evolution starting from dislocation multiplication leading to planar slip activity followed by slip band refinement with the applied strain in the fully recrystallized and precipitate-free austenite grains of all the alloys. In the deformed austenite regions of partially recrystallized sample (HR + CR+973 K), fine cross slip activity resulting into formation of dislocation cell structure was seen (Fig. 11a). In regions of post-deformed austenite containing coarser and fine κ-carbides there were certain differences in slip band evolution. Dislocations were cutting through fine κ-carbides evolving into slip band structure with a mean slip band spacing of ~200 ± 30 nm. Further, dislocation-twin interaction was observed and is shown in Fig. 11e. In case of the alloys containing coarse and high fraction of κ -carbides along with B2 particles refinement of slip band spacing was delayed at earlier stages of deformation, which resulted in inhomogeneous slip band spacing. Non-shearable B2 particles were also noticed as obstacles to dislocation motion and thereby early initiation of slip band refinement. In summary, inhomogeneous slip band spacing was developed in Ni containing LDS alloys, whereas Ni-free LDS (28Mn0Ni) alloy exhibited homogeneous slip band spacing under similar heat-treated conditions of HR + CR+1173 K.

4. Discussion

In the present study three low-density austenitic steels of 28Mn0Ni, 28Mn5Ni and 15Mn5Ni were chosen in the as-hot rolled condition. This was with an intention to investigate the response of initial microstructure, which was a resultant of the composition, to the cold reduction as

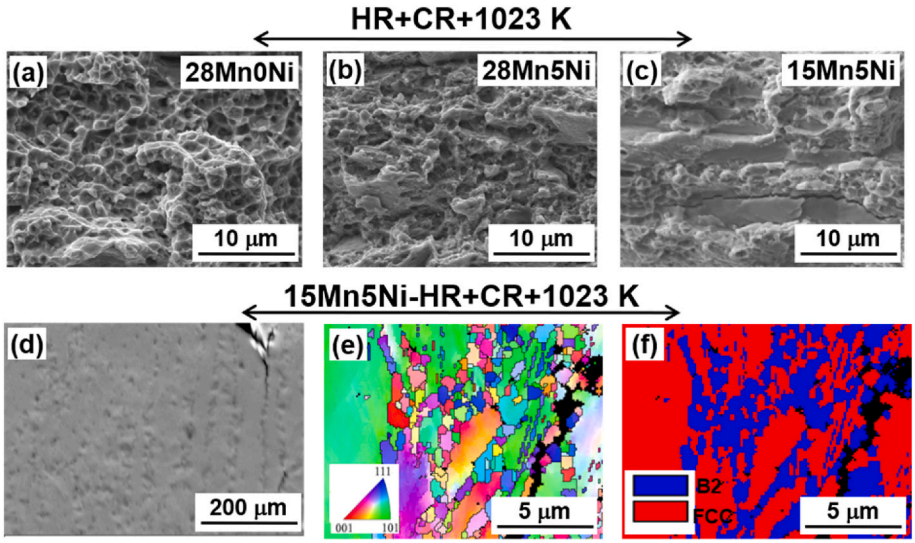


Fig. 10. SEM-fractographs of the failed tensile specimens of (a) 28Mn0Ni, (b) 28Mn5Ni and (c) 15Mn5Ni alloys in HR + CR+1023 K condition. (d) SEM image, (e) OIM-IPF map and (f) phase contrast map of cross-section of failed tensile specimen of 15Mn5Ni alloy in same heat-treated condition.

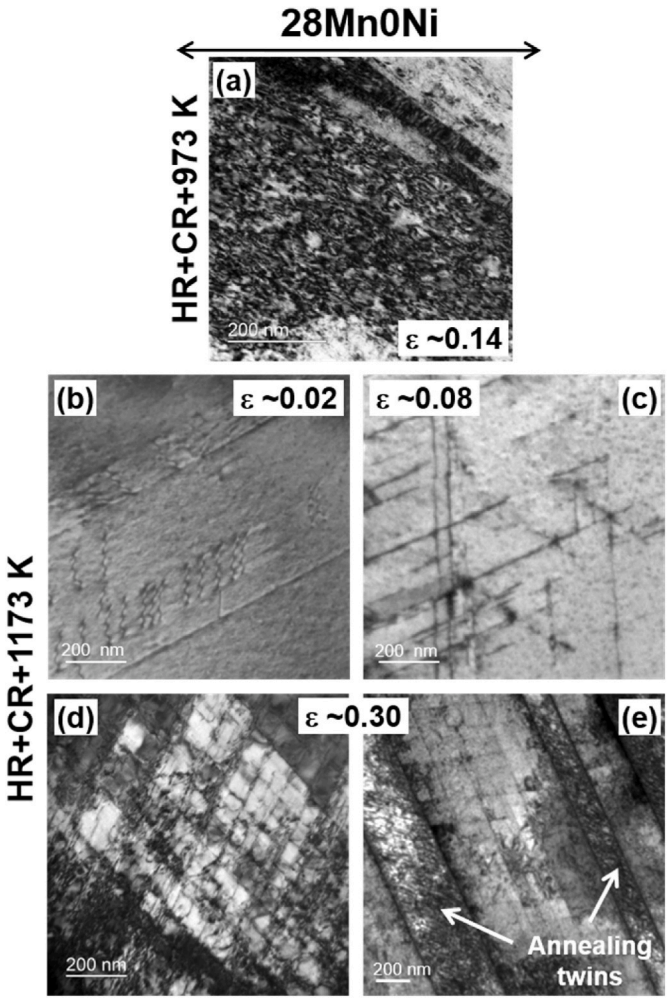


Fig. 11. TEM images showing post tensile deformed microstructures of 28Mn0Ni alloy in (a) HR + CR+973 K condition deformed to a true strain, ε~0.14, (b–e) HR + CR+1173 K condition deformed to ε~0.02, 0.08 and 0.30. Here, (a) shows dislocation multiplication and cell formation, (b–d) show refinement of slip bands with progress of deformation (strain), and (e) indicates interaction between dislocation and prior annealing twins.

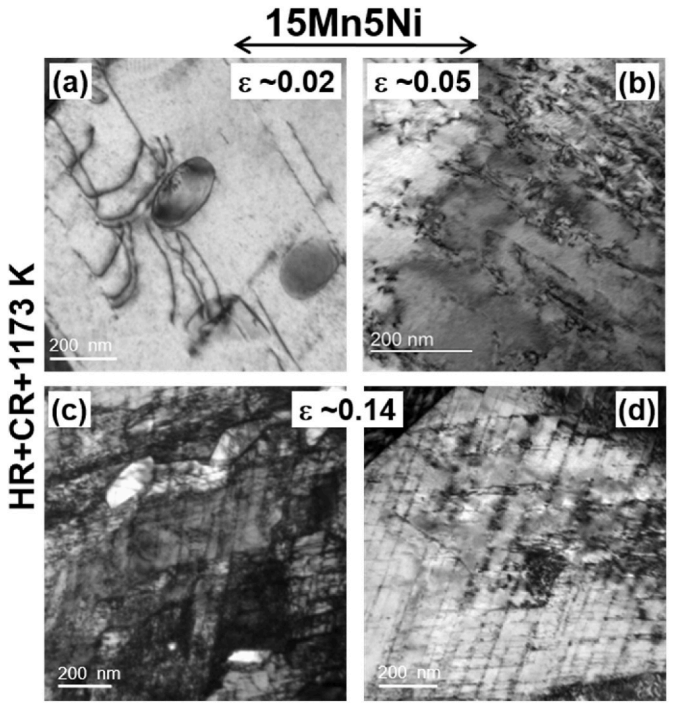


Fig. 12. TEM images showing post tensile deformed microstructures of 15Mn5Ni alloy in HR + CR+1173 K condition deformed to ε~0.02, 0.05 and 0.14. Here, (a) indicates dislocation-B2 particle interaction and (b–d) show dislocation multiplication and refinement of slip bands with strain.

well as subsequent heat treatment and eventually to evaluate their mechanical behavior in both cold rolled and heat-treated conditions. All the three alloys could be rolled successfully without any cracking at room temperature till 50% reduction in thickness. As anticipated, there was an increased hardness, but at a differential work hardening rate in all the three alloys with the degree of cold reduction. The strength of 50% cold rolled alloys increased significantly to ~1.4–1.8 GPa but at the expense of their ductility. This is very much expected in metallic materials, as there can be an increased dislocation density with the amount of cold working and thereby, enhanced strengthening owing to Taylor hardening ($\sigma \propto \rho^{1/2}$). It was also observed that rate of increase in hardness with progress of deformation after 30% reduction in thickness decreased significantly. The hardness of all the three alloys at 50% reduction in thickness was almost similar. This differential work hardening rate with cold work as a function of alloy composition could entirely be related to the prior microstructural features such as austenite grain size and B2 phase volume fraction present in the respective alloys. In our recent study [21] it was noted the selected alloys in hot rolled condition exhibited about 20–70% of total tensile elongation to failure at room temperature, where 15Mn5Ni alloy displayed lowest ductility. This suggested that there will be an increased tendency for cracking when the amount of cold rolling exceeds 50% reduction in thickness. Furthermore, the recent literature [22,23] indicated that when the cold reduction is beyond 50% the austenitic LDS alloys undergo severe shear bands formation, which can affect the mechanical properties adversely. Hence, the cold rolling was restricted to 50% of reduction in thickness for all the three alloys for obtaining a better comparison of the microstructural evolution in the chosen alloys for a given degree of cold work and subsequently to evaluate their mechanical properties.

Recently, some researchers have attempted to enhance the strength-ductility tradeoff in Ni-free LDS by processing the material at lower temperature range (873 K–1073 K) and thus controlling the interplay between κ-carbide size/fraction and recovery/recrystallization process [18,19]. This was in contrast to the earlier studies where strength enhancement in LDS alloys was attempted in fully recrystallized austenite matrix by precipitation hardening. In case of Ni containing LDS alloys, a novel strategy of dual precipitation of κ-carbides and Ni rich B2 precipitates showed pathways to obtain higher strength-ductility combination [31,32]. These studies have opened up a new possibility of property enhancement in lightweight induced plasticity (L-IP) steels by combination of multiple strengthening mechanisms such as shear band/microband induced plasticity, dynamic slip band refinement, deformation induced precipitation hardening apart from conventional precipitation hardening by either κ-carbide and/or B2 phase. However, the role of alloy composition and corresponding hardening/softening mechanism haven't yet been reported. The present study delved into fulfill this gap by highlighting the role of alloy composition *vis-à-vis* the heat treatment temperature on these mechanisms. The synergistic effects of Mn and Ni on concurrent precipitation and recrystallization have been addressed in the following sections.

4.1. Role of alloy composition on concurrent precipitation and recrystallization

Recrystallization in metallic alloys is affected by the presence of secondary phase in two ways depending on size of the second phase. Coarse second phase particles are known to promote particle simulated

nucleation (PSN), whereas fine particles will induce Zener-Smith drag effect [33]. In the present alloy systems, stringer B2 can act as PSN site and aids in recrystallization in Ni-containing alloys. The finer intergranular/intragranular B2/κ-carbide precipitation could occur concurrently during heat treatment. The size of the precipitate changes during thermal treatment and thereby affects the recrystallization in a complex way. The competing processes of concurrent B2/κ-carbide precipitation and recrystallization kinetics can have profound effect on final microstructure, which in turn depends on the alloy composition and heat treatment temperature/time. Figs. 3, 4 and 6 clearly revealed that nucleation of recrystallized grains occurs mostly along the γ/γ boundaries and γ/B2 interface. It can be inferred that the intragranular B2 particles will not pin the newly recrystallized grains rather may delay the overall recovery process.

It was well established that LDS alloys containing Al >7% (in wt.%) and C >0.7% (in wt.%) exhibit κ-carbides precipitation at elevated temperatures [12,34]. Size and dissolution of κ-carbides, however, depend upon alloying and ageing temperature-time combination. In the present study, relatively coarser κ-carbides precipitation was observed within the deformed austenite regions, which was consistent with the results of Zhang et al. [17] and Brasche et al. [19]. It was suggested that these κ-carbides form by spinodal decomposition during quenching [35,36]. It was noted that lowering Mn content in presence of Ni has pronounced effect on coarsening of κ-carbides than just Ni addition in high Mn LDS. Dual precipitation of κ-carbides and B2 was also observed in 28Mn5Ni and 15Mn5Ni alloys and is consistent with the earlier study [32]. Nickel rich B2 is known to favor increased rejection of carbon in the neighborhood austenite and thereby facilitates κ-carbides formation in Ni added LDS as compared to Ni-free LDS [32]. It seems that 28Mn5Ni alloy has higher austenite stability and lesser B2 precipitation and thus, limiting rejection of carbon around B2. Very fine κ-carbides precipitation was observed within recrystallized grains and is agreement with literature [29,30,37,38]. These precipitates promote glide plane softening and accordingly, higher rate of strain hardening [29,30,39].

It was observed that with increase in heat treatment temperature from 973 K to 1173 K, the volume fraction of B2 particles decreased drastically from ~40% to ~15% in 15Mn5Ni alloy, while its fraction was too reduced but slightly in the other two alloys. It is to be noted that the total volume fraction of B2 consisted of three different types include stringer, intragranular and intergranular. It is important to note that the volume fraction was calculated by including type-1 and type-3 B2 particles characterized at EBSD scale and was found to be much higher in 15Mn5Ni alloy. However, there was high volume fraction of type-2 i.e., fine intragranular B2 particles, noticed even in 28Mn5Ni alloy at TEM scale. Dissolution rate of fine intragranular as well as intergranular B2 was noted to be higher than that of B2 stringers. This resulted into substantial decrease in volume fraction of B2 in 15Mn5Ni alloy. The presence of higher Mn in 28Mn5Ni alloy led to less B2 precipitation at all heat treatment temperatures. The high rate of reduction in volume fraction of B2 in 15Mn5Ni alloy with increase in temperature resulted in reversal in recrystallization volume fraction of austenite at 1173 K. As noted in Figs. 5a and 28Mn5Ni alloy showed lesser recrystallization volume fraction at 1173 K, while it was almost similar in both 15Mn5Ni and 28Mn5Ni alloys up to 1023 K. This indicates pronounced concurrent B2 precipitation effect on recrystallization kinetics as compared to κ-carbides. Furthermore, there was no noticeable growth of recrystallized grains in Ni-containing LDS alloys up to 1173 K. Thus, it can be inferred that in presence of secondary phase the recrystallization phenomenon can be delayed in Ni containing LDS alloys.

It is well-known that Mn and C are austenite stabilizers, whereas Al is a ferrite stabilizer in steels. Though Ni is known to stabilize austenite, the high (negative) enthalpy of formation between Ni and Al can result into formation of Ni rich B2 particles during heat treatment. The absence of Ni enhanced recrystallization and grain growth kinetics in 28Mn0Ni alloy significantly, whereas concurrent B2 precipitation in 28Mn5Ni alloy delayed recrystallization of austenite due to change in diffusion characteristics and boundary migration. At similar Ni levels, change in Mn content (28Mn5Ni and 15Mn5Ni) affected stability of austenite grain structure as well as composition and volume fraction of B2 phase, and thus affecting solubility of precipitates and recrystallization of austenite.

It has been well established that heat treatment in the two-phase region controls grain growth in most metallic systems because of reduced grain boundary migration and mobility due to decreased diffusivities. This is the case in the present study as well. For example, as can be noted in Fig. 4b,d,f and Fig. 5b that when the alloys were heat treated at 1173 K significant austenite grain growth was noticed in 28Mn0Ni alloy (B2-free alloy), but not in the other two alloys containing secondary B2 phase. This clearly suggests that heat treatment in the two-phase region helps in attaining refined grain structure in LDS alloys as well.

4.2. *Effect of concurrent precipitation and recrystallization on tensile properties*

In the present study, 28Mn0Ni alloy in HR + CR+1173K condition exhibited strength levels of YS~500 MPa, UTS~1000 MPa and total elongation ~86% and is consistent with prior annealing studies in single phase region [4–7]. Attempts were made to tailor the microstructure of 28Mn0Ni alloy to enhance strength-ductility combination by thermo-mechanical treatment in two-phase region i.e., at temperature of 973 K. There was significant increase in yield strength to ~900 MPa, along with a reasonable ductility of ~23% tensile elongation, without any additional ageing treatment. This can be attributed to the bimodal microstructure involving the combined effects of deformed austenite with intragranuar κ-carbides causing strengthening, while the recrystallized grains contribute to softening. Recrystallization kinetics below 973 K was found to be too sluggish and hence, requires a prolonged heat treatment time. However, an increase in the heat treatment temperature by 50 K i.e., at 1023 K, accelerated the recrystallization kinetics but at the expense of drastic reduction in strength. Thus, it is inferred that the processing temperature window for 28Mn0Ni alloy is around 973 K. Further fine tuning the heat treatment temperature and/or time around 973 K may enhance the strength-ductility combination in this alloy, but is beyond scope of the present study.

In view of accelerated recrystallization kinetics at and above 1023 K, where the recrystallization fraction was >90% in Ni-free LDS leading to significant drop in strength, Ni addition was explored to facilitate Ni rich B2 formation and thereby to control recrystallization at higher temperatures. This idea was derived primarily from literature, for example, Ni addition was first suggested by Kim et al. [5] that resulted into remarkable increase in strength without much loss in ductility (20–30%) in Fe–15Mn–9Al–5Ni alloy. Higher strength was attributed to non-shearable Ni-rich B2 particles. In our recent study [21] it was noted that B2 in stringer morphology is the limiting factor for ductility. In the present study too, it was observed that upon heat treatment at 973 K and 1023 K, 15Mn5Ni alloy displayed very low ductility (<1%). This could be entirely related to the presence of large volume fraction of B2 particles and co-precipitation of coarser κ-carbides and B2 along grain boundaries. Thus, the processing temperature window for 15Mn5Ni alloy was found to be around 1173 K, which can yield optimum combination of strength and ductility.

Recent studies have indicated that Ni addition in higher Mn containing LDS alloy i.e., Fe–29Mn–9Al-1.2C–6Ni can result into better strength-ductility combination by heat treatment at 1073 K–1173 K [32]. The properties were found to be superior to the earlier Ni containing LDS at lower Mn levels. In the present study, unlike in 28Mn0Ni alloy, there was just 10% recrystallization in 28Mn5Ni alloy. Dual precipitation of intragranular B2, intergranular B2 and κ-carbides along with partially recrystallized structure led to better strength (1.3–1.6 GPa) and ductility (17%) combination in 28Mn5Ni alloy. With further increase in the heat treatment temperature to 1173 K, there was

substantial drop in strength (YS = 900 MPa, UTS = 1250 MPa) with corresponding increase in ductility (39%) in 28Mn5Ni alloy. This could be due to change in size of B2 precipitates and concurrent changes in recrystallization kinetics. Thus, the thermo-mechanical processing window to attain better strength-ductility combination in 28Mn5Ni alloy should involve tailoring microstructure so as to contain fine nanosized precipitates.

It is noted that the higher volume fraction of deformed austenite grains along with κ-carbides were present in low temperature heat treated sample as compared to that observed in the samples heat treated at 1173 K (Figs. 6 and 7), and this can alter the deformation mechanisms. High strain hardening ability of fully recrystallized LDS alloys could be associated with the planar slip activity followed by dynamic slip band refinement process (Figs. 11 and 12). Similar observations pertaining to high strain hardening rate and associated slip band refinement were also shown in various earlier studies [5–8]. In addition, dislocation-twin interaction associated with dynamic slip band refinement was found to be the dominant strain hardening mechanism in recrystallized alloys. Lower strain hardening at 973 K in 28Mn0Ni alloy can be attributed to higher volume fraction of deformed austenite region containing coarse κ-carbides. As noticed from TEM examination (Figs. 6 and 7) that the austenite grains contained inhomogeneous distribution of κ-carbides and B2 particles in heat-treated conditions. The presence of these particles and their uneven size distribution can influence the change of deformation mechanism from planar glide to wavy slip involving dislocation cross slip (at local scale) owing to the intrinsic high SFE of the alloys. Further, dislocation substructure evolution in these regions suggested that dynamic slip band refinement may be impeded in such deformed grains at local scale. Moreover, the limited ductility in low temperature heat treated samples of Ni containing LDS alloys can be related not only to the reduced strain hardening ability but also to crack-initiation controlled second phase particles presence.

5. Conclusions

Three different low-density steels, namely Fe–28Mn–9Al-0.9C (28Mn0Ni), Fe–28Mn–9Al-0.9C–5Ni (28Mn5Ni) and Fe–15Mn–9Al-0.9C–5Ni (15Mn5Ni) were used in the present study to understand the role of concurrent microstructural changes and their effect on thermo-mechanical processing window. The role of Mn/Ni (ratio) on dual precipitation of κ-carbides as well as B2 phase, which in turn their interplay with dislocation induced hardening/softening were analyzed. The alloy composition was chosen such a way that there is a varying B2 phase fraction from 0 to ~12% in hot rolled condition. The significant findings of the present study are summarized as follows:

- Upon cold rolling up to 50% reduction in thickness, the austenite grains in all three alloys deformed mostly by planar slip as reflected by parallel slip bands, while the prior B2 stringers too underwent concurrent progressive microstructural changes ranging from dislocation accumulation, substructure development and fragmentation. As anticipated, the mechanical strength and hardness of these alloys increased with the degree of cold reduction at the expense of their ductility. All the three alloys attained similar strength levels after 50% cold reduction with the maximum rate of work hardening in a single-phase LDS i.e., 28Mn0Ni alloy.
- Subsequent heat treatment after 50% cold rolling resulted in substantial microstructural changes such as recrystallization and grain growth phenomena in austenite along with precipitation and dissolution of secondary phases (κ-carbides and B2 phase) depending on the temperature and composition of the alloy. For a given heat treatment condition i.e., at and above 973 K, 28Mn0Ni alloy underwent higher rate of recrystallization and grain coarsening in conjunction with increased twin fraction than that of the other two Ni-containing alloys.
- Mechanical property enhancement in Ni-free LDS (28Mn0Ni alloy) by thermo-mechanical processing is controlled by dissolution of κ-carbides. At temperatures less than the dissolution temperature of κ-carbides, deformed zone containing κ-carbides has led to the strength enhancement but limited ductility. A slight heat treatment temperature rise by 50 K has affected the recrystallization fraction of austenite matrix grains significantly and accordingly, influenced the strength-ductility combination in Ni-free LDS.
- Ni addition in high Mn LDS (28Mn5Ni alloy) delayed recrystallization significantly. Furthermore, Ni addition along with thermo-mechanical processing were found to be promising due to concurrent fine precipitates of κ-carbides as well as Ni rich B2 phase in two-phase LDS. This facilitates wider processing window for optimizing the mechanical properties.
- Ni containing lower Mn LDS (15Mn5Ni alloy) had detrimental effect of larger volume fraction of B2 (including prior B2-stringers) and κ-carbides precipitation under low heat treatment temperatures of 973 K and 1023 K i.e., in two-phase region. Upon heat treatment at higher temperature of 1173 K, complete dissolution of κ-carbides and partial dissolution of B2 led to sudden rise in recrystallization fraction of austenite grains. Similar strength-ductility levels along with improved strain hardening characteristics were noticed in both Ni containing alloys upon high temperature heat treatment at 1173 K.

Credit author statement

Bidyapati Mishra: Methodology, Investigation, Formal analysis, Writing – original draft, **Rajdeep Sarkar**: Investigation, Validation, **Vajinder Singh**: Investigation, Validation, **Deepak Kumar**: Investigation, Validation, **A. Mukhopadhyay**: Resources, Supervision, **V. Madhu**: Resources, Supervision. **M.J.N.V. Prasad**: Conceptualization, Methodology, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

Authors are grateful to the Director, DMRL for his encouragement and giving permission for publishing this work and acknowledge DRDO for funding. Authors would like to acknowledge the members of various research groups such as Special Melting Group (SMG), Rolling Forging Group (RFG) and Armour Technology Center (ATC) in DMRL for their help rendered in preparation of the steel grades.

References

[1] S. Chen, R. Rana, A. Haldar, R.K. Ray, Current state of Fe-Mn-Al-C low density steels, Prog. Mater. Sci. 89 (2017) 345–391, https://doi.org/10.1016/j.pmatsci.2017.05.002.
[2] H. Kim, D.W. Suh, N.J. Kim, Fe-Al-Mn-C lightweight structural alloys: a review on the microstructures and mechanical properties, Sci. Technol. Adv. Mater. 14 (2013), https://doi.org/10.1088/1468-6996/14/1/014205, 014205 1-01420511.
[3] M.S. Chen, H.C. Cheng, C.F. Huang, Chao, K.L. Ou, C.H. Yu, Effects of C and Cr content on high temperature microstructures of Fe-9Al-30Mn-xC-yCr alloys, Mater. Char. 61 (2010) 206–211, https://doi.org/10.1016/j.matchar.2009.11.011.
[4] I. Gutierrez-Urrutia, Low density Fe-Mn-Al-C steels: phase structures, mechanism and properties, ISIJ Int. 61 (2021) 16–25, https://doi.org/10.2355/isijinternational.ISIJINT-2020-467.
[5] S.H. Kim, H. Kim, N.J. Kim, Brittle intermetallic compound makes ultra strong low-density steel with large ductility, Nature 518 (2015) 77–79, https://doi.org/10.1038/nature14144.

[6] A. Rahnama, H. Kotadia, S. Sridhar, Effect of Ni alloying on the microstructural evolution and mechanical properties of two duplex light-weight steels during different annealing temperatures: experiment and phase-field simulation, Acta Mater. 132 (2017) 627–643, https://doi.org/10.1016/j.actamat.2017.03.043.
[7] A. Rahnama, H. Kotadia, S. Clark, V. Janik, S. Sridhar, Nano-mechanical properties of Fe-Mn-Al-C lightweight steels, Sci. Rep. 6 (2018) 9065, https://doi.org/10.1038/s41598-018-27345-w.
[8] A. Rahnama, S. Sridhar, Multiscale micromechanical model of duplex low density steel with B2 ordered precipitates, Mech. Mater. 131 (2019) 136–140, https://doi.org/10.1016/j.mechmat.2019.02.003.
[9] M.X. Yang, F.P. Yuan, Q.G. Xie, Y.D. Wang, E. Ma, X.L. Wu, Strain hardening in Fe-16Mn-10Al-0.86C-5Ni high specific strength steel, Acta Mater. 109 (2016) 213–222, https://doi.org/10.1016/j.actamat.2016.02.044.
[10] J.H. Hwang, T.T.T. Trang, O. Lee, G. Park, A. Zargaran, N.J. Kim, Improvement of strength-ductility balance of B2-strengthened lightweight steel, Acta Mater. 191 (2020) 1–12, https://doi.org/10.1016/j.actamat.2020.03.022.
[11] M.J. Yao, E. Wesch, D. Ponge, S.M.H. Haghighat, S. Sadlobes, P. Choi, M. Herbig, I. Bleskov, T. Hickel, M. Lipinska-Chawlek, P. Santhraj, C. Scheu, S. Zaefferer, B. Gault, D. Raabe, Strengthening and strain hardening mechanism in a precipitation-hardened high-Mn lightweight steel, Acta Mater. 140 (2017) 258–273, https://doi.org/10.1016/j.actamat.2017.08.049.
[12] M.C. Li, H. Chang, P.W. Kao, D. Gan, The effect of Mn and Al contents on the solvus of κ phase in austenitic Fe-Mn-Al-C alloys, Mater. Chem. Phys. 59 (1999) 96–99, https://doi.org/10.1016/S0254-0584(99)00026-7.
[13] K. Sato, K. Tagawa, Y. Inoue, Modulated structure and magnetic properties of age-hardenable Fe-Mn-Al-C alloys, Metall. Trans. A 21 (1990) 5–11, https://doi.org/10.1007/BF02656419.
[14] W. Song, W. Zhang, J.V. Appen, R. Dronskowski, W. Bleck, κ-Phase Formation in Fe-Mn-Al-C austenitic steels, Steel Res. Int. 86 (10) (2015) 1161–1169, https://doi.org/10.1002/srin.201400587.
[15] I. Gutierrez-Urrutia, D. Raabe, Influence of Al content and precipitation state on the mechanical behavior of austenitic high-Mn low-density steels, Scripta Mater. 68 (2013) 343–347, https://doi.org/10.1016/j.scriptamat.2012.08.038.
[16] C.L. Lin, C.G. Chao, H.Y. Bor, T.F. Liu, T.F. Liu, Relationship between microstructures and tensile properties of Fe-30Mn-8.5Al-2C alloy, Mater. Trans. 51 (2010) 1084–1088, https://doi.org/10.2320/matertrans.M2010013.
[17] J. Zhang, D. Raabe, C.C. Tasan, Designing duplex, ultrafine-grained Fe-Mn-Al-C steels by tuning phase transformation and recrystallization kinetics, Acta Mater. 141 (2017) 374–387, https://doi.org/10.1016/j.actamat.2017.09.026.
[18] G. Zhang, W. Ma, Y. Tang, F. Wang, X. Zhang, Q. Wang, R. Liu, Investigation on the microstructural evolution and mechanical properties of partially recrystallized Fe-27Mn-10Al-1.4C steel, Mater. Sci. Eng. 833 (2022), 142545, https://doi.org/10.1016/j.msea.2021.142545.
[19] F. Brasche, C. Haase, M.L. Chwalek, J. Mayer, D.A. Molodov, Combined κ-carbide precipitation and recovery enables ultra-high strength and ductility in light-weight steels, Mater. Sci. Eng. 795 (2020) 139928, https://doi.org/10.1016/j.msea.2020.139928.
[20] S.W. Hwang, J.H. Ji, E.G. Lee, K.T. Park, Tensile deformation of a duplex Fe–20Mn–9Al–0.6C steel having the reduced specific weight, Mater. Sci. Eng. 528 (2011) 5196–5203, https://doi.org/10.1016/j.msea.2011.03.045.
[21] B. Mishra, R. Sarkar, V. Singh, A. Mukhopadhyay, R.T. Mathew, V. Madhu, M.J.N. V. Prasad, Microstructure and deformation behaviour of austenitic low-density steels: the defining role of intermetallic B2 phase, Materialia 20 (2021), 101198, https://doi.org/10.1016/j.mtla.2021.101198.
[22] P. Ren, X.P. Chen, C.Y. Wang, Y.X. Zhou, W.Q. Cao, Q. Liu, Evolution of microstructure, texture and mechanical properties of Fe–30Mn–11Al–1.2C low-density steel during cold rolling, Mater. Char. 174 (2021), 111013, https://doi.org/10.1016/j.matchar.2021.111013.
[23] F. Ji, W. Song, Y. Ma, C. Li, W. Bleck, G. Wang, Recrystallization behavior in a low-density high-Mn high-Al austenitic steel undergone thin strip casting process, Mater. Sci. Eng. 733 (2018) 81–97, https://doi.org/10.1016/j.msea.2018.07.023.
[24] M.C. Ha, J.M. Koo, J.K. Lee, S.W. Hwang, K.T. Park, Tensile deformation of a low density Fe–27Mn–12Al–0.8C duplex steel in association with ordered phase at ambient temperature, Mater. Sci. Eng. 586 (2013) 276–283, https://doi.org/10.1016/j.msea.2013.07.094.
[25] Z. Huang, A. Hou, Y. Ziang, P. Wang, Q. Shi, Q. Hou, X. Liu, Rietveld refinement, microstructure, mechanical properties and oxidation characteristics of Fe-28Mn-xAl-1C (x = 10-12 wt. %) low-density steels, J. Iron Steel Res. Int. 24 (2007) 1190–1198, https://doi.org/10.1016/S1006-706X(18)30017-7.
[26] J.B. Seol, A brief review of κ-carbide in Fe-Mn-Al-C model alloys, Appl. Micros. 48 (2018) 117–121, https://doi.org/10.9729/AM.2018.48.4.117.
[27] E. Welsch, D. Ponge, S.M.H. Haghighat, S. Sandlöbes, P. Choi, M. Herbig, S. Zaefferer, D. Raabe, Strain hardening by dynamic slip band refinement in a high-Mn lightweight steel, Acta Mater. 116 (2016) 188–199, https://doi.org/10.1016/j.actamat.2016.06.037.
[28] H. Kim, Strain hardening of novel high Al low-density steel consisting of austenite matrix and B2-ordered intermetallic second phase in the perspective of non-cell forming face-centered-cubic alloy with high stacking fault energy, Scripta Mater. 160 (2019) 29–32, https://doi.org/10.1016/j.scriptamat.2018.09.044.
[29] C. Haase, C. Zehnder, T. Ingendahl, A. Bikar, F. Tang, B. Hallstedt, W. Hu, W. Bleck, D.A. Molodov, On the deformation behavior of κ-carbide-free and κ-carbide-containing high-Mn light-weight steel, Acta Mater. 122 (2017) 332–343, https://doi.org/10.1016/j.actamat.2016.10.006.
[30] I. Gutierrez-Urrutia, D. Raabe, Multistage strain hardening through dislocation substructure and twinning in a high strength and ductile weight-reduced Fe-Mn-Al-C steel, Acta Mater. 60 (2012) 5791–5802, https://doi.org/10.1016/j.actamat.2012.07.018.
[31] A. Zargaran, T.T.T. Trang, G. Park, N.J. Kim, κ-carbide assisted nucleation of B2: a novel pathway to develop high specific strength steel, Acta Mater. 220 (2021), 117349, https://doi.org/10.1016/j.actamat.2021.117349.
[32] Z. Wang, W. Lu, H. Zhao, C.H. Leibscher, J. He, D. Ponge, D. Raabe, Z. Li, Ultrastrong light weight compositionally complex steels via dual-nanoprecipitation, Sci. Adv. 6 (2020) 1–7, https://doi.org/10.1126/sciadv.aba9543.
[33] J. Humphrey, G.S. Rohrer, A. Rollett, Chapter 9- Recrystallization of two–phase alloys, in: Recrystallization and Related Annealing Phenomena, third ed., Elsevier, 2017, pp. 321–359, https://doi.org/10.1016/B978-0-08-098235-9.00009-4.
[34] H. Huang, D. Gan, P.W. Kao, Effect of alloying additions on the κ phase precipitation in austenitic Fe-Mn-Al-C alloys, Scripta Metall. Mater. 30 (1994) 499–504, https://doi.org/10.1016/0956-716X(94)90610-6.
[35] K. Sato, K. Tagawa, Y. Inoue, Spinodal decomposition and mechanical properties of an austenitic Fe-30wt.%Mn-9wt.%Al-0.9wt.%C alloy, Mater. Sci. Eng. 111 (1989) 45–50, https://doi.org/10.1016/0921-5093(89)90196-2.
[36] W.C. Cheng, C.Y. Cheng, C.W. Hsu, D.E. Laughlin, Phase transformation of the L_{12} phase to kappa-carbide after spinodal decomposition and ordering in an Fe–C–Mn–Al austenitic steel, Mater. Sci. Eng. 642 (2015) 128–135, https://doi.org/10.1016/j.msea.2015.06.096.
[37] J.H. Shin, T.H. Lee, Direct observation of dislocation plasticity in high-Mn lightweight steel by in-situ TEM, Sci. Rep. 9 (2019), 15171, https://doi.org/10.1038/s41598-019-51586-y.
[38] J.D. Yoo, K.T. Park, Micro-band induced plasticity in a high Mn-Al-C light steel, Mater. Sci. Eng. 496 (2008) 417–442, https://doi.org/10.1016/j.msea.2008.05.042.
[39] V. Gerold, H.P. Kranthaler, On the origin of planar slip in f.c.c. alloys, Acta Metall. 37 (1989) 2177–2183, https://doi.org/10.1016/0001-6160(89)90143-0.